

Analysis PhD Exam, May 2001

I

State the following theorems:

1. The Hahn–Banach theorem.
2. The monotone class theorem.
3. The monotone convergence theorem in $L^1(\mu)$.
4. The Lebesgue dominated convergence theorem in $L^p(\mu)$.
5. The density of step functions in $L^p(\mu)$, $1 \leq p < \infty$, and $L^\infty(\mu)$.
6. The computation of the seminorm $\|f\|_p$.
7. The integrability with respect to a measure defined by density.
8. The Fubini theorem for $\mu \times \nu$ -integrable functions.
9. The Lusin theorem.
10. The Riesz representation theorem.

II

State and prove the following theorems:

1. The Egorov theorem.
2. The integral representation of the dual of $L^1(\mu)$, μ finite.

III

Solve the following problems:

1. Prove that if X is a normed space and Y is a Banach space, then $L(X, Y)$ is a Banach space.
2. Prove the existence of the conditional expectation for functions $f \in L^1(\mu)$ and for functions $f \in L^1_{\mathcal{F}}(\mu)$.
3. Prove that the set of step functions is dense in $L^\infty(\mu)$.
4. Let (X, Σ, μ) be a measure space and $\mathcal{R}$ a ring generating the σ -algebra Σ . Let $g : X \rightarrow \mathbb{R}$ be a μ -measurable function. Prove that if $\int_A g d\mu = 0$ for every $A \in \mathcal{R}$, then $g = 0$, μ -a.e. Is this true for $g : X \rightarrow F$, where F is a Banach space?
5. Let (X, Σ) be a measurable space and (μ_n) a sequence of real valued measures on Σ . Define $\lambda : \Sigma \rightarrow \mathbb{R}_+$ by

$$\lambda(A) = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{|\mu_n|(A)}{1 + |\mu_n|(X)}, \quad A \in \Sigma.$$

Prove that λ is finite, σ -additive on Σ , and $\mu_n \ll \lambda$ for each n .

Note: Write complete computations and explanations. Do not use arrows or other unnecessary signs. Use words for explanations. Write complete sentences. State the requisite formulae for each theorem and problem.